Testing LakeTemperature

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Project Goals

- Develop a software that can systematically change the constants used in the LakeTemperature model and allow us to visualize and observe patterns in how this affects its output.
- Apply testing techniques to try and find inputs that will cause physically impossible or unrealistic.

LakeTemperature Model Interface

- Is a unit test produced by SPEL for the LakeTemperature module with a specific dataset.
- Allows us to change the values of constants normally specified in LakeCon using the file lakeparams.txt.
- Provides a file cpu_LakeTemperatureTest.txt that lists output values for comparison purposes.
- We can set values in lakeparams.txt, run elmtest.exe, and get results inside cpu_LakeTemperatureTest.txt.

Our Testing Solution

- Automates the process of inputting constants into lakeparams.txt and the process of analyzing the resulting values in cpu_LakeTemperatureTest.txt.
- Both the process of picking values for the constants and the process of analyzing the resulting outputs is extendable through sampling plugins and analysis plugins respectively.

What is a Sample?

- A sample is a set of values for the constants contained in lakeparams.txt.
- elmtest.exe is run once for every sample.

lakeparams.txt

```
betavis
0.0
za_lake
0.6
n2min
7.4999999999999993E-005
tdmax
277.0
pudz
0.0
mixfact
10.0
depthcrit
25.0
```

Example Sample

```json
{
  "betavis": 0.0,
  "za_lake": 0.6,
  "n2min": 7.4999999999999993E-005,
  "tdmax": 277.0,
  "pudz": 0.0,
  "mixfact": 10.0,
  "depthcrit": 25.0
}
```

What Outputs are Received per Sample?

- Analysis plugins will receive the outputs from the cpu_LakeTemperatureTest.txt file in the form of a dictionary.

cpu_LakeTemperatureTest.txt

```
 veg_ef%eflx_soil_grnd
   1.0000000000000000E+036   1.0000000000000000E+036   1.0000000000000000E+036
   1.0000000000000000E+036    77.5917042259827 8        1.0000000000000000E+036
  veg_ef%eflx_gnet
   1.0000000000000000E+036   1.0000000000000000E+036   1.0000000000000000E+036
   1.0000000000000000E+036    76.9873654621642 5        1.0000000000000000E+036
```

Outputs Received by Analysis Plugins


```
{
  "veg_ef%eflx_soil_grnd": [
    1.0000000000000000E+036, 1.0000000000000000E+036, 1.0000000000000000E+036,
    1.0000000000000000E+036, 77.59170422598278, 1.0000000000000000E+036
  ],
  "veg_ef%eflx_gnet": [
    1.0000000000000000E+036, 1.0000000000000000E+036, 1.0000000000000000E+036,
    1.0000000000000000E+036, 76.98736546216425, 1.0000000000000000E+036
  ],
}
```